Ong Lab Take Home Report

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Question

In Neuro ICU patients, do pupillometry measurements help predict neurologic deterioration beyond what we know at admission?

1. Clean and describe the data.

600 simulated patients and about 28,000 nurse pupil checks. Every decision and its reason is in out/cleaning_log.csv. To summarize how data was cleaned, removed 138 duplicate exam rows, set 169 left-eye NPi values of -99 to missing, divided 25 right-eye NPi values of 20–49 by 10 and recoded 18 inconsistent sex codes. As for the missing data, 9.4% of exams could not be done, and 89% of patients had at least one failed exam, which could mean it is not random. Failed exams were twice as common in the 12 h before deterioration, and more common with sedation, low admission GCS and prior cataract surgery. Therefore, pupil values are carried forward from the last valid exam, and "last exam could not be done" is used as a predictor.

2. Survival analysis

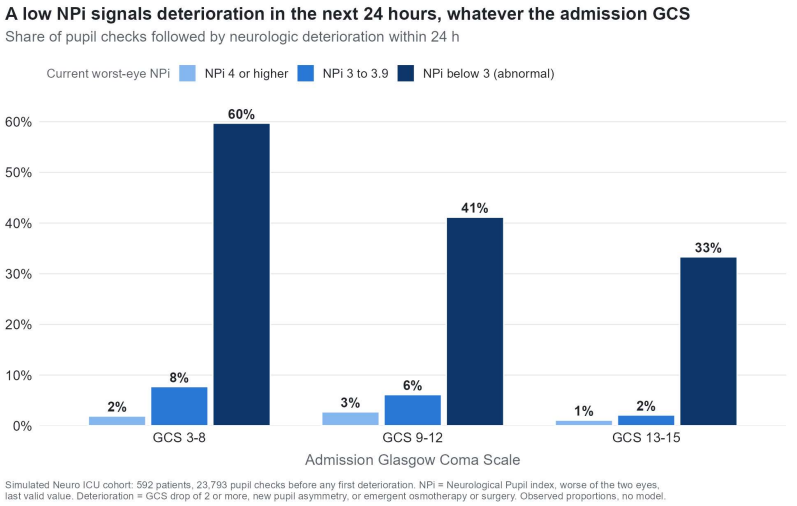
Cox model with pupil values updated at every exam, using only exams that happened before each moment. The approach to repeated exams and their timing is explained at the top of 02_survival.R. 129 of 600 patients (22%) deteriorated, at a median of 42 hours after admission. Each 1-point fall in worst-eye NPi carried about an 8-fold higher hazard of deterioration (HR 8.2, 95% CI 5.3–12.8), adjusted for all admission variables. Left–right NPi asymmetry and a check that could not be done each added risk too, and once NPi was known, admission GCS and midline shift no longer added much. Adding pupillometry to admission data raised the C-index from 0.73 to 0.98. Results held when pupil data were delayed by 2 h. Prior cataract surgery lowers NPi for eye reasons, so a low NPi in these patients was less ominous (based on only 30 patients).

3. Prediction model

A random forest that, at every pupil check, estimates the chance of deterioration in the next 24 hours. The split (70/30) and framing are explained at the top of 03_prediction.R. It reached a test AUROC of 0.81, against 0.70 for admission data alone (difference 0.12, 95% CI 0.05 to 0.20), with good calibration: on the test set, calibration intercept was 0.05 and slope 1.05 (ideal 0 and 1). Flagging checks with predicted risk of 20% or more would flag about 5% of checks, catch about a third of upcoming events, and be right about 4 times in 10.

4. Figure

The figure shows that after a check with NPi below 3, 33–60% of patients deteriorated within 24 hours, against 1–3% when NPi was 4 or higher, across every admission GCS group.



Conclusion

Yes, pupillometry measurements help predict neurological deterioration beyond what we know at admission. In this simulated cohort, adding pupillometry to admission data improved 24-hour prediction (AUROC 0.81 vs 0.70).

References

Therneau, "Using Time Dependent Covariates and Time Dependent Coefficients in the Cox Model" (survival vignette); Therneau & Grambsch, Modeling Survival Data: Extending the Cox Model (Springer); Wright & Ziegler, "ranger" (Journal of Statistical Software, 2017); Robin et al., "pROC" (BMC Bioinformatics, 2011); Van Calster et al., "Calibration: the Achilles heel of predictive analytics" (BMC Medicine, 2019); Steyerberg, Clinical Prediction Models (Springer, 2nd ed.); Zeileis, "Object-Oriented Computation of Sandwich Estimators" (Journal of Statistical Software, 2006); Chen et al., "Pupillary reactivity as an early indicator of increased intracranial pressure: the introduction of the Neurological Pupil index" (Surgical Neurology International, 2011).